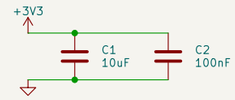
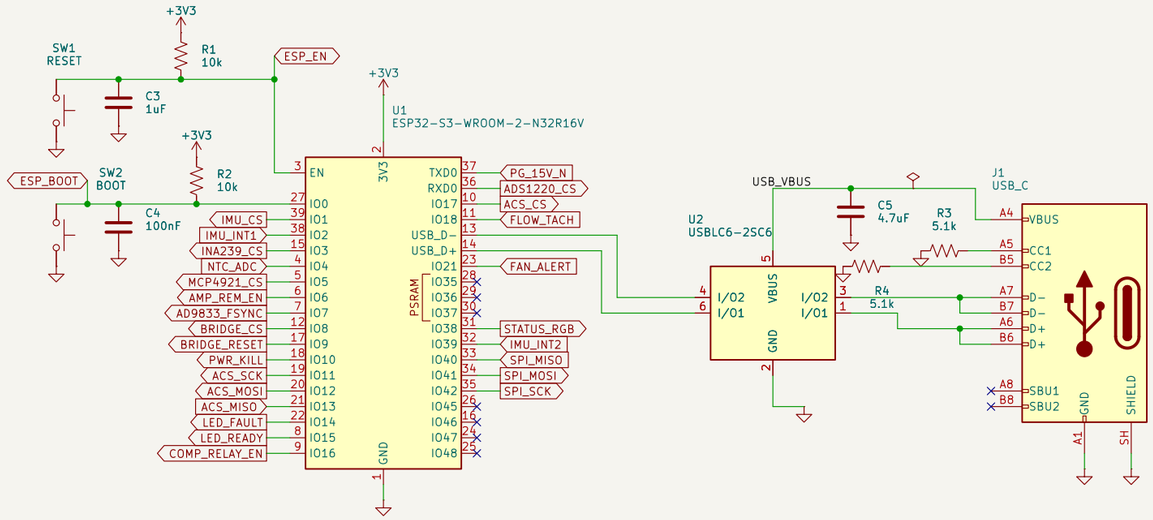
ESP32-S3-WROOM-2-N32R16V: 32 MB octal flash + 16 MB octal PSRAM (GPIO26-37 reserved by module)

GPIO map follows include/config/pin_config.h

ESP32-S3 DECOUPLING CAPACITORS





Power Input
File: sheets/power_input.kicad_sch

Signal Generator
File: sheets/signal_gen.kicad_sch

Water Cooling
File: sheets/cooling.kicad_sch

Power Button
File: sheets/power_button.kicad_sch

Amp Control
File: sheets/amp_control.kicad_sch

Cold Head Temp
File: sheets/cold_head.kicad_sch

Rails 5V 3V3
File: sheets/rails_logic.kicad_sch

Amp Current
File: sheets/amp_current.kicad_sch

IMU
File: sheets/imu.kicad_sch

Rails +-15V
File: sheets/rails_15v.kicad_sch

Power Good
File: sheets/power_good.kicad_sch

System Current
File: sheets/system_current.kicad_sch

Indicators
File: sheets/indicators.kicad_sch

Compressor Relay
File: sheets/compressor_relay.kicad_sch

Water Cooling I2C Bridge
File: sheets/i2c_bridge.kicad_sch

Debug
File: sheets/debug.kicad_sch

Supply Switches

Cryocooler Controller

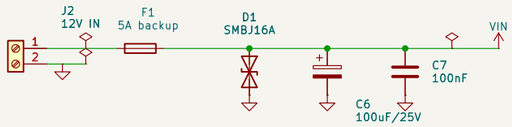
Sheet: /
File: cryocooler.kicad_sch

Title: Cryocooler Controller

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/18

12 V input from a Dell XYK93 (DPS-2000EB class, ~12.2 V, ~160 A available): use the main 12 V rail with PS_ON held low.

F1 is a backup only: add an inline high-breaking-capacity fuse (7.5-10 A) at the PSU end. Reverse polarity: TVS forward-clamps and blows the fuse.



Cryocooler Controller

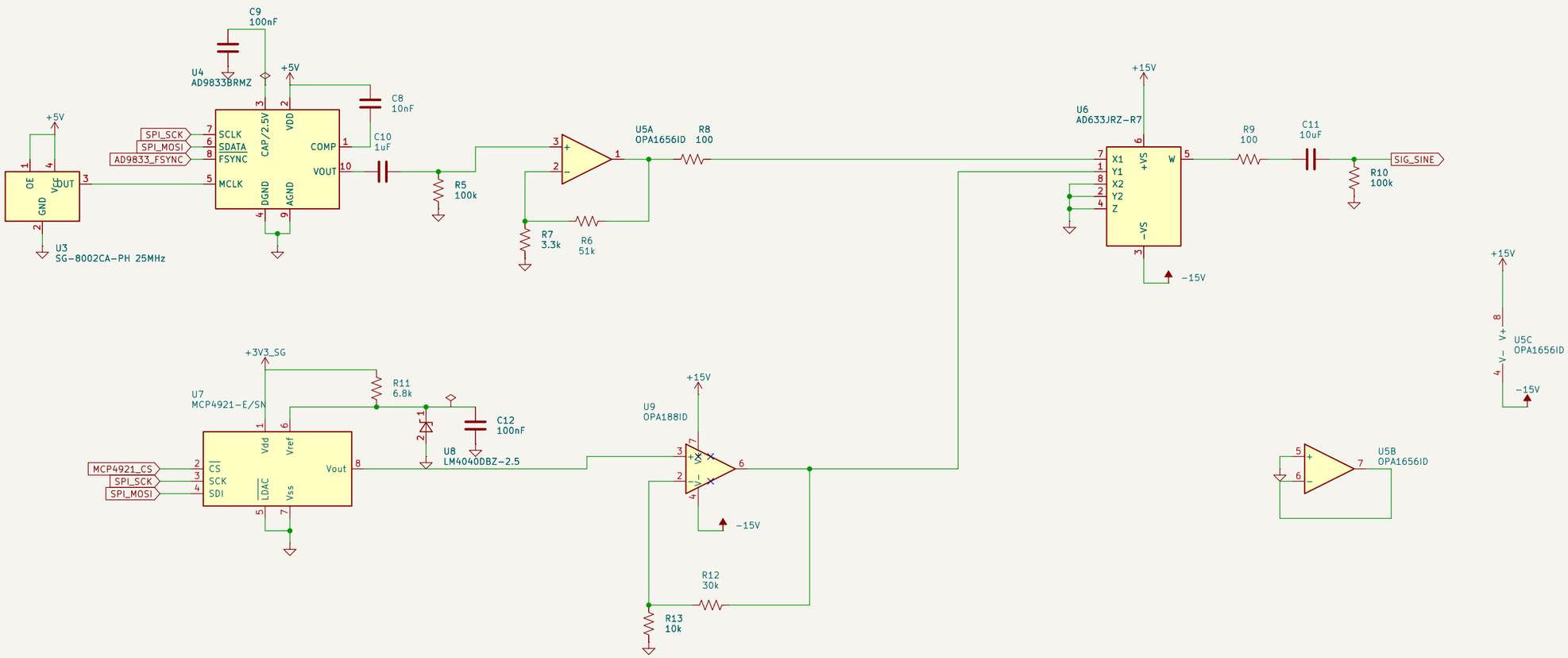
Sheet: /Power Input/
File: power_input.kicad_sch

Title: Power Input

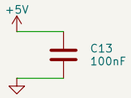
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/18

AD9833 60 Hz sine -> OPA1656 gain stage -> AD633 multiplier; MCP4921 + OPA188 give a 0–10 V amplitude control.

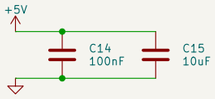
Vctl 0–10 V (2.5 V ref x4) x sine +/–4.9 V pk / 10 = 0..4.9 V pk (9.9 Vpp, 3.5 Vrms) to amplifier input; ZE500.1 RCA input range 0.2–6 V.



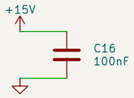
SG–8002CA DECOUPLING CAPACITORS



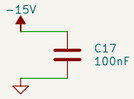
AD9833 DECOUPLING CAPACITORS



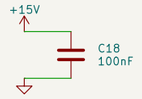
OPA1656 +15V DECOUPLING CAPACITORS



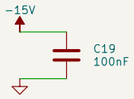
OPA1656 –15V DECOUPLING CAPACITORS



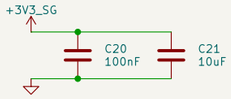
AD633 +15V DECOUPLING CAPACITORS



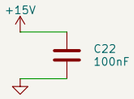
AD633 –15V DECOUPLING CAPACITORS



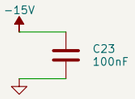
MCP4921 DECOUPLING CAPACITORS



OPA188 +15V DECOUPLING CAPACITORS



OPA188 –15V DECOUPLING CAPACITORS



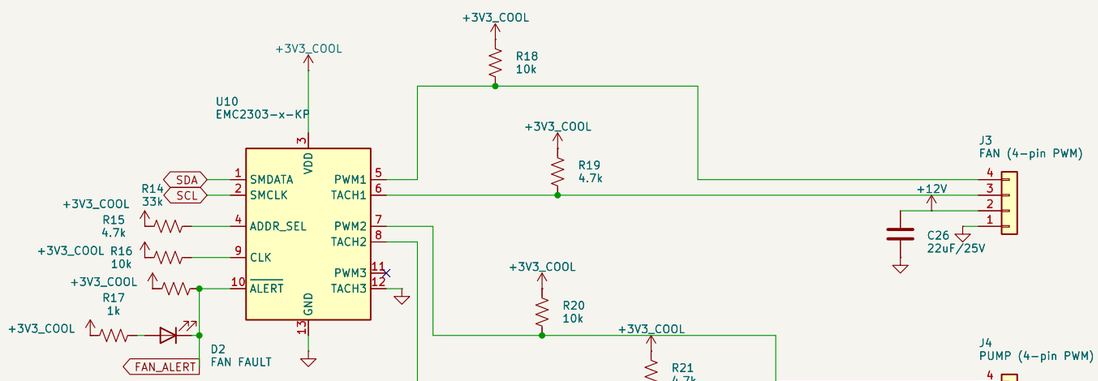
Cryocooler Controller

Sheet: /Signal Generator/
File: signal_gen.kicad_sch

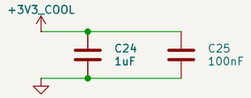
Title: Signal Generator

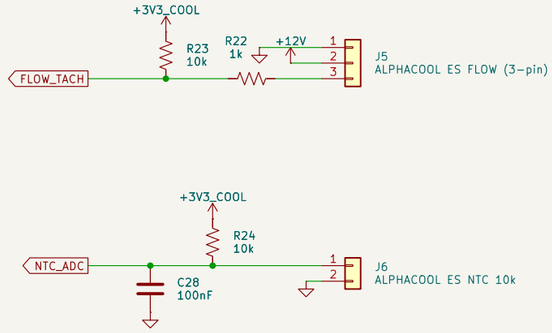
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 3/18

EMC2303: ADDR_SEL 33k pull-up -> SMBus 0x4D (datasheet Table 5-1); CLK 4.7k pull-up -> default fan drive 0%. ch1 fans, ch2 pump. Alphacool ES: tach -> GPIO18, NTC -> GPIO4.



EMC2303 DECOUPLING CAPACITORS





Cryocooler Controller

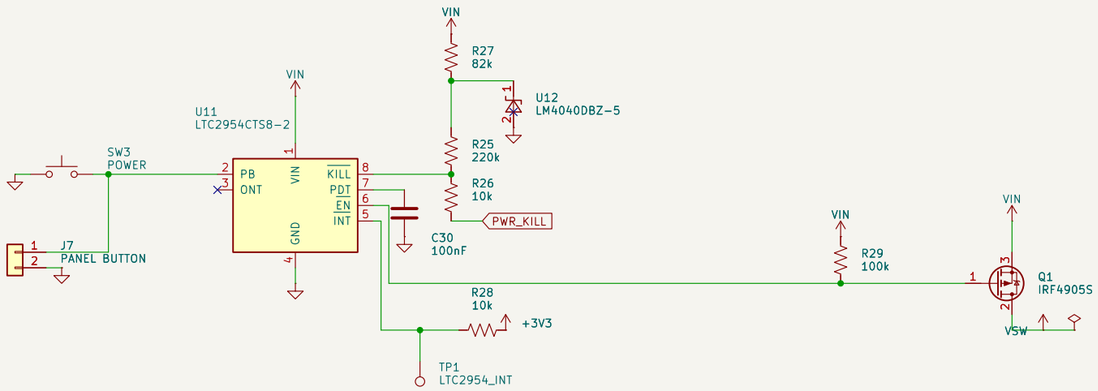
Sheet: /Water Cooling/
File: cooling.kicad_sch

Title: Water Cooling

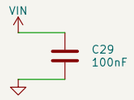
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 4/18

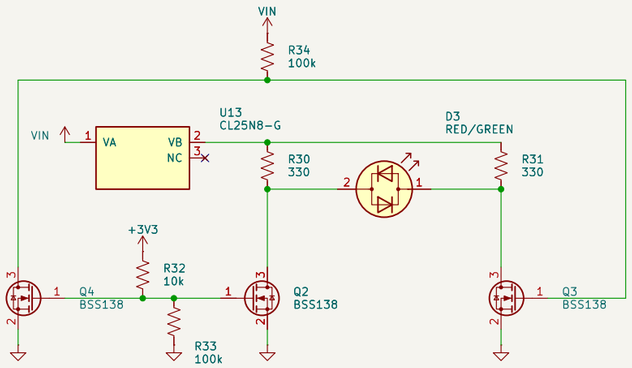
LTC2954-2: tap PB = on, hold = off. EN# drives the P-FET gate; KILL# from GPIO10 (active low, 100 ms pulse).

PWR_ON follows the +3V3 rail: on = green, standby = red. CL25 (25 mA) feeds both legs, ~9 mA reaches the LED, ~25 mA total from VIN in standby



LTC2954 DECOUPLING CAPACITORS

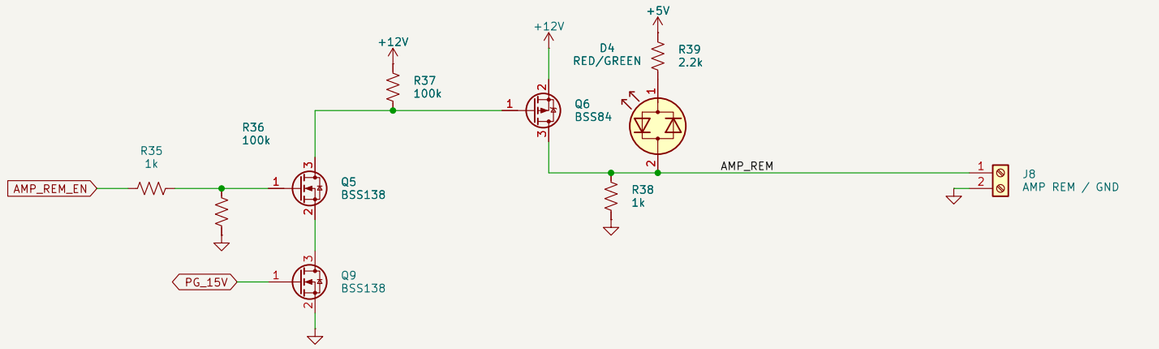




Cryocooler Controller		
Sheet: /Power Button/		
File: power_button.kicad_sch		
Title: Power Button		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 5/18

GPIO6 high -> +12 V on the amplifier REM terminal (P-FET high-side switch), but only while PG_15V is high (both +-15 V rails good: Q9 in series with the level shifter). LED: green = on, red = off.

Amplifier: AUDIOZERONE ZE500.1 (external): REM terminal + RCA/line input. Its power leads do not pass through this board.





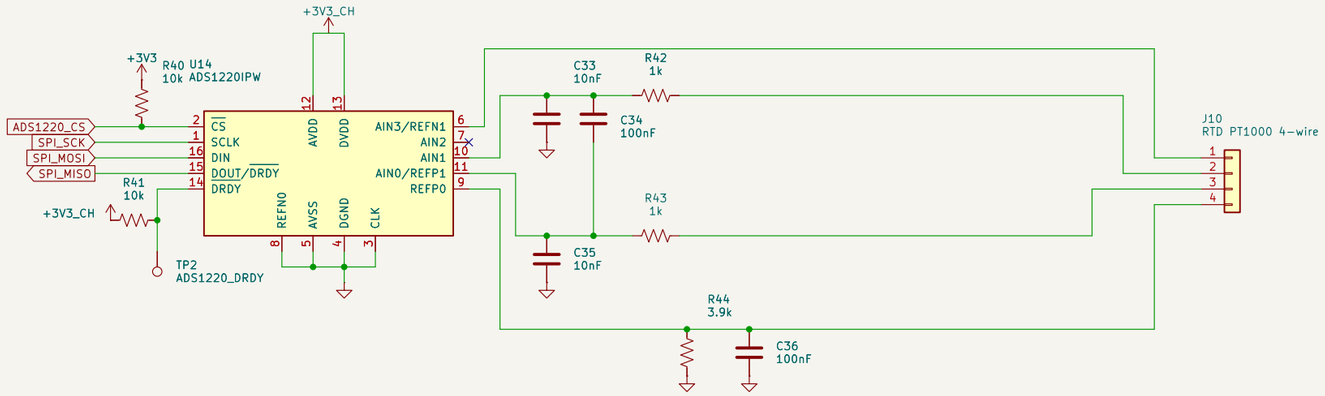
Cryocooler Controller

Sheet: /Amp Control/
File: amp_control.kicad_sch

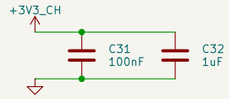
Title: Amp Control

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 6/18

PT1000 4-wire ratiometric: IDAC1 (AIN3) excites, AIN1/AIN0 sense, Rref (3.9k) on REFP0/REFN0. SPI mode 1 on the main bus (CS GPIO44), internal oscillator (CLK tied low). DRDY is not wired to the MCU.



ADS1220 DECOUPLING CAPACITORS



Cryocooler Controller

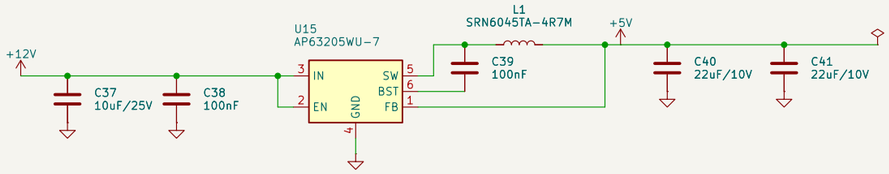
Sheet: /Cold Head Temp/
File: cold_head.kicad_sch

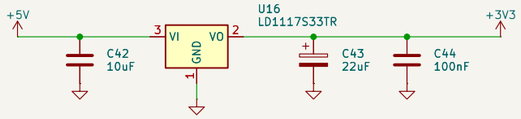
Title: Cold Head Temp

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 7/18

12 V -> 5 V (AP63205 buck, datasheet Fig. 1 / Table 3: 4.7 uH, 10 uF in, 2 x 22 uF out, 100 nF bootstrap) -> 3.3 V (LD1117S33).

AP63205: VIN 3.8–32 V (40 V for 400 ms), EN tied to VIN, FB tied to the fixed 5 V output.





Cryocooler Controller

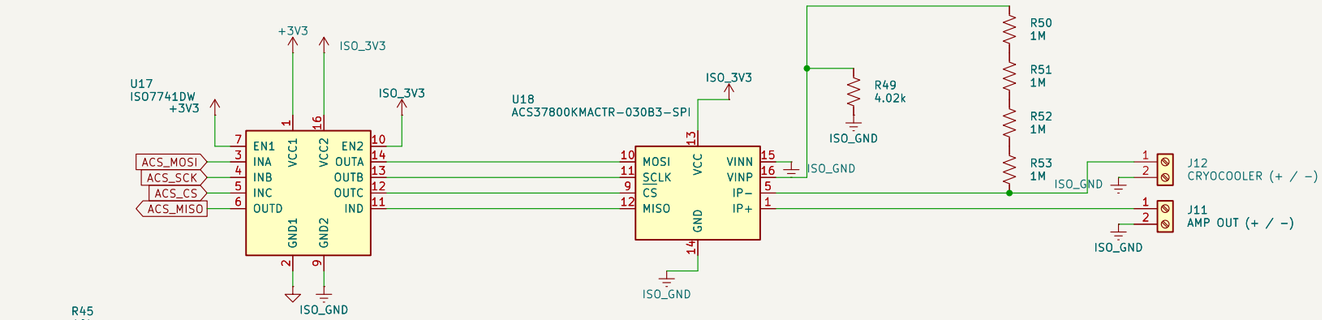
Sheet: /Rails 5V 3V3/
File: rails_logic.kicad_sch

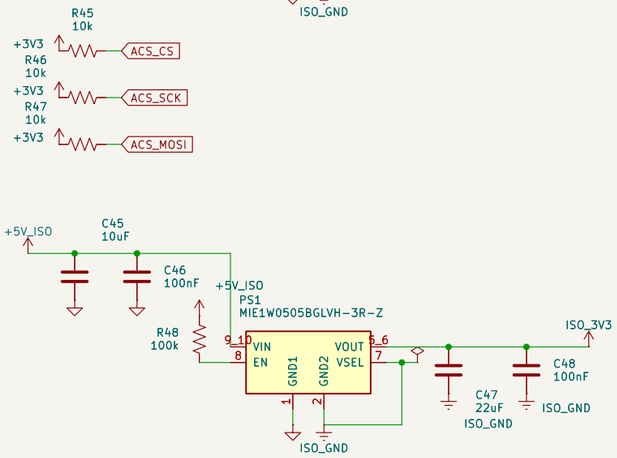
Title: Rails 5V 3V3

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 8/18

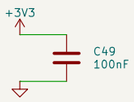
Amplifier output V/I monitor: ACS37800 (SPI) on an isolated 3.3 V domain (MIE1W0505, VSEL=GND2 -> 3.3 V); SPI bus GPIO11/12/13, CS GPIO17.

Mains-class creepage/clearance applies between ISO_GND domain and the logic side on the PCB.

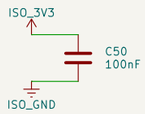




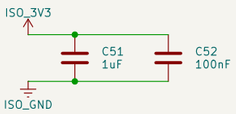
ISO7741 VCC1 DECOUPLING CAPACITORS



ISO7741 VCC2 DECOUPLING CAPACITORS



ACS37800 DECOUPLING CAPACITORS



Cryocooler Controller

Sheet: /Amp Current/

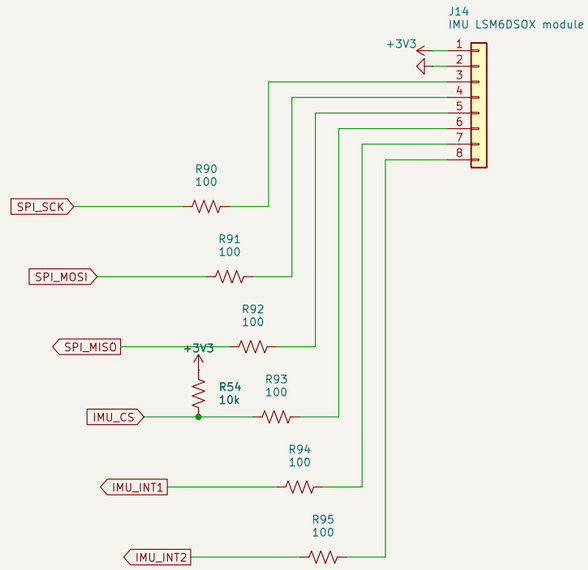
File: amp_current.kicad_sch

Title: Amp Current

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 9/18

LSM6DSOX is the off-board Adafruit module (product 4438) on the cooler, cabled to J14 over the main SPI bus (CS GPIO1, INT1 GPIO2 single-tap, INT2 GPIO39 wake-up).

Module Vin = +3V3 keeps its logic at 3.3 V (its 3Vo pad stays open). 100 R series resistors damp the cable stub; the 10 k keeps CS high with the cable unplugged.



Cryocooler Controller

Sheet: /IMU/
File: imu.kicad_sch

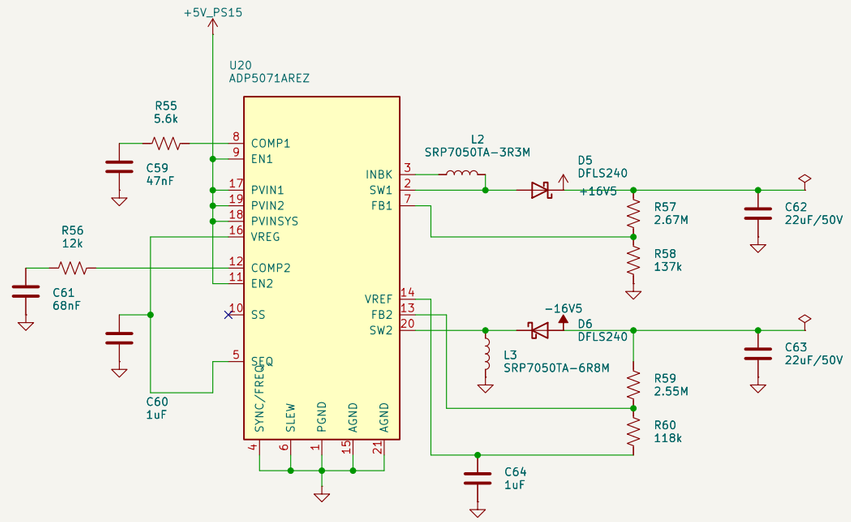
Title: IMU

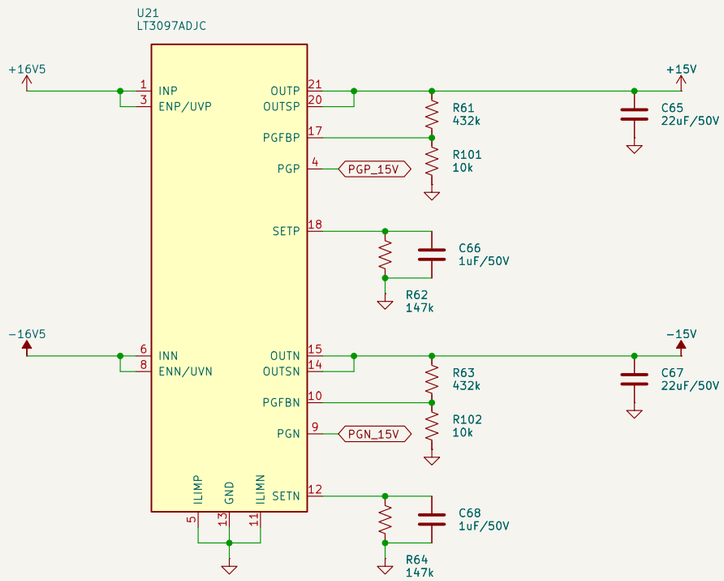
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 10/18

ADP5071AREZ (5 V in): +16.4 V boost and −16.5 V inverting rails; one LT3097 post-regulates them to +/−14.7 V.

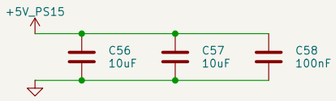
ADP5071 values from datasheet Fig. 47 / Tables 9–11. LT3097: VOUT = 100 uA x RSET (147k = 14.7 V, +15 V is the top of its range); PGFB dividers 432k/10k trip power good near 13.3 V (also enables fast start-up).

PGP and PGN (open collector, low = bad) go to nodes PGP_15V / PGN_15V built on the power_good sheet, which also makes PG_15V and lights one LED per pin.





ADP5071 DECOUPLING CAPACITORS



Cryocooler Controller

Sheet: /Rails +−15V/
File: rails_15v.kicad_sch

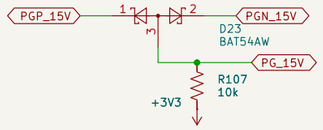
Title: Rails +−15V

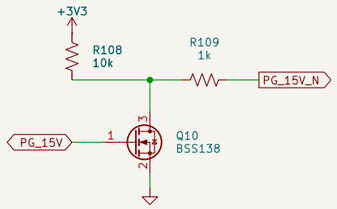
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 11/18

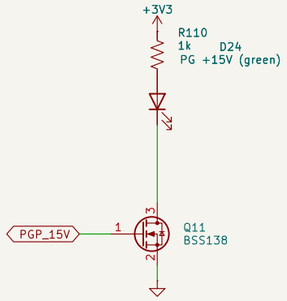
PGP and PGN pull their own node low when bad. Each node is the +15 V rail through 220k/62k (about 3.2 V when good, 0 V if its pin pulls low or the rail is missing) and lights its green LED.

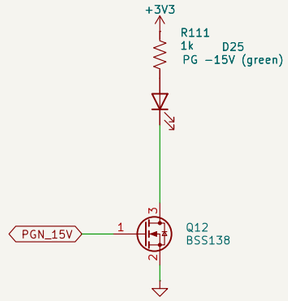
PG_15V = the two nodes ANDed by a BAT54AW with a 10k pull-up to +3V3 (3.3 V only when both are good); it gates the amplifier REM switch and reads LOW on GPIO43 (PG_15V_N) when both rails are good.











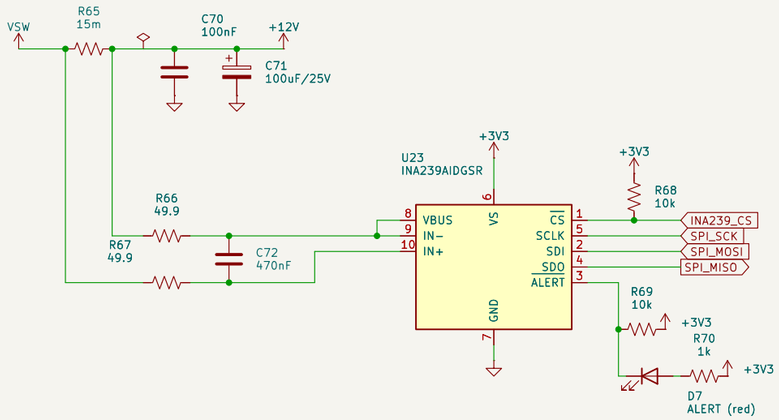
Cryocooler Controller

Sheet: /Power Good/
File: power_good.kicad_sch

Title: Power Good

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 12/18

INA239 monitors the switched 12 V rail: 15 mOhm shunt, SPI mode 1 on the main bus (CS GPIO3), +/−163.84 mV range (~10.9 A). ALERT only drives its LED.



3V3 POWER LED

INA239 DECOUPLING CAPACITORS

Cryocooler Controller

Sheet: /System Current/
File: system_current.kicad_sch

Title: System Current

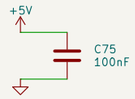
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 13/18

READY (GPIO15) and FAULT (GPIO14) LEDs, active high; WS2812B status LED on GPIO38 via a 74AHCT1G125 3.3 V -> 5 V level shifter (WS2812B VIH = 0.7 x VDD).

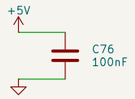


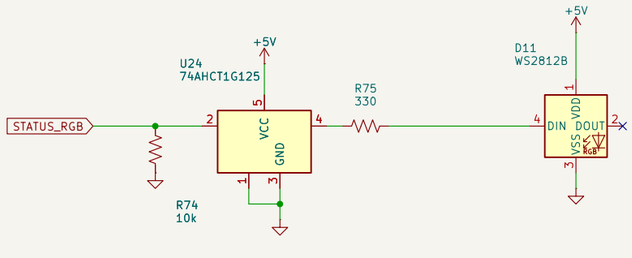


74AHCT1G125 DECOUPLING CAPACITORS



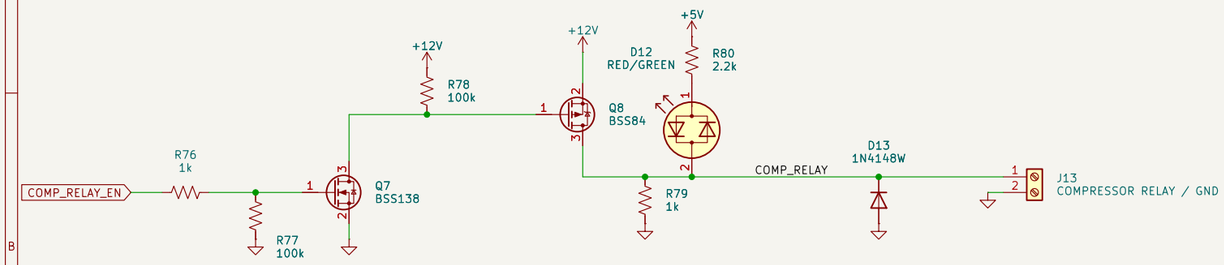
WS2812B DECOUPLING CAPACITORS





Cryocooler Controller		
Sheet: /Indicators/ File: indicators.kicad_sch		
Title: Indicators		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 14/18

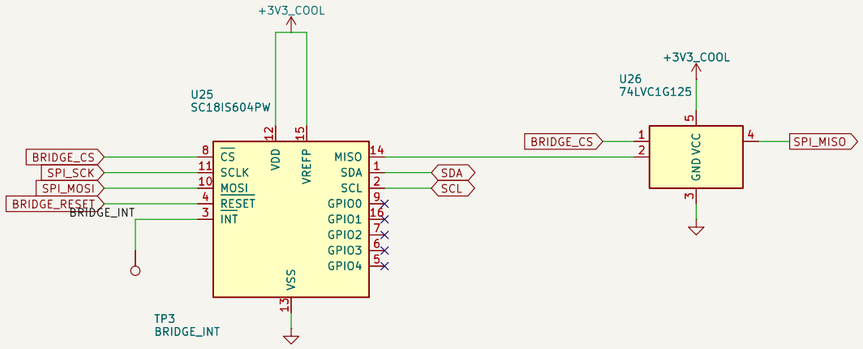
COMP_RELAY_EN (GPIO16) high -> +12 V on the compressor relay terminal (BSS84 high-side switch, 130 mA max: drive an external relay or SSR coil). Flyback diode across the output. LED: green = on, red = off.



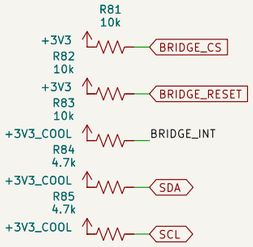
Cryocooler Controller		
Sheet: /Compressor Relay/ File: compressor_relay.kicad_sch		
Title: Compressor Relay		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 15/18

SC18IS604: SPI mode 3 slave on the main bus (CS GPIO8, RESET GPIO9), I2C controller at ~100 kHz for the EMC2303 (0x4D). INT# is polled through the I2CStat register and only has a pull-up and a test point.

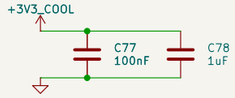
The datasheet does not say MISO is high-impedance with CS high, so a 74LVC1G125 (OE# = CS) gates it onto SPI_MISO.



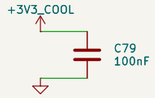
PULL-UPS



SC18IS604 DECOUPLING CAPACITORS



74LVC1G125 DECOUPLING CAPACITORS



Cryocooler Controller

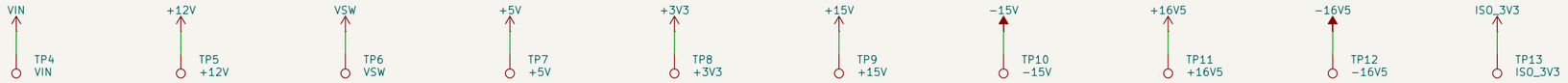
Sheet: /Water Cooling I2C Bridge/
File: i2c_bridge.kicad_sch

Title: Water Cooling I2C Bridge

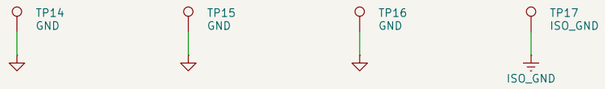
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 16/18

Debug aids: test points (pad per net, loops for ground) and rail LEDs. Each test point sits on a global label or power symbol; the PCB generator places it beside a pad on the same net.

POWER RAILS



GROUND (scope loops)



SPI BUSES AND CHIP SELECTS



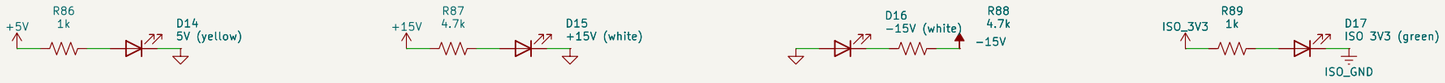
I2C, INTERRUPTS AND CONTROL



ANALOG AND SENSORS



RAIL LEDS



Cryocooler Controller

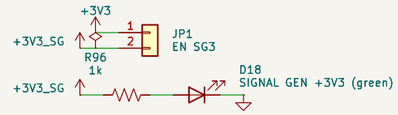
Sheet: /Debug/
File: debug.kicad_sch

Title: Debug

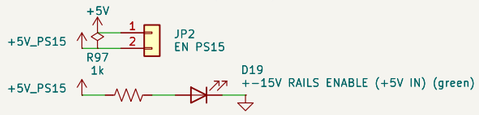
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 17/18

Each 1.27 mm jumper sits in series with one load supply: pull it to power that part off. The LED shows the switched rail is present. A part left unpowered can still be fed through its signal pins, so disconnect the cable too when you need it fully dead.

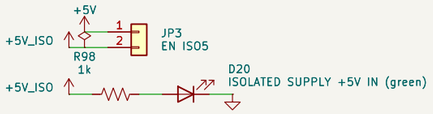
SIGNAL GEN +3V3



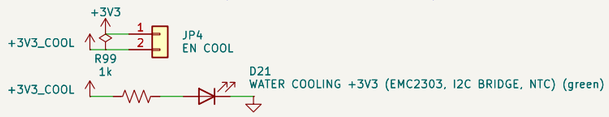
+−15V RAILS ENABLE (+5V IN)



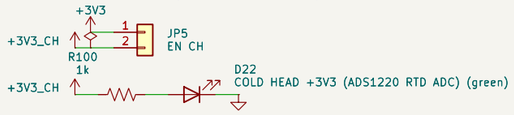
ISOLATED SUPPLY +5V IN



WATER COOLING +3V3 (EMC2303, I2C BRIDGE, NTC)



COLD HEAD +3V3 (ADS1220 RTD ADC)



Cryocooler Controller

Sheet: /Supply Switches/
File: supply_switches.kicad_sch

Title: Supply Switches

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 18/18